



TARGET-90

JKBOSE CLASS-11TH

TRIGONOMETRIC FUNCTION

One shot

MATHS

Mehraj sir



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$



TOPICS TO BE COVERED

- Introduction to Trigonometry
- Degree and radian Measure
- T Functions as Circular functions
- Various Trigonometric identities

(Important Questions
+ PYQ's)

Miscellaneous exercise
 ↳ Tigo ↳ Complex ↳ Seq. Series

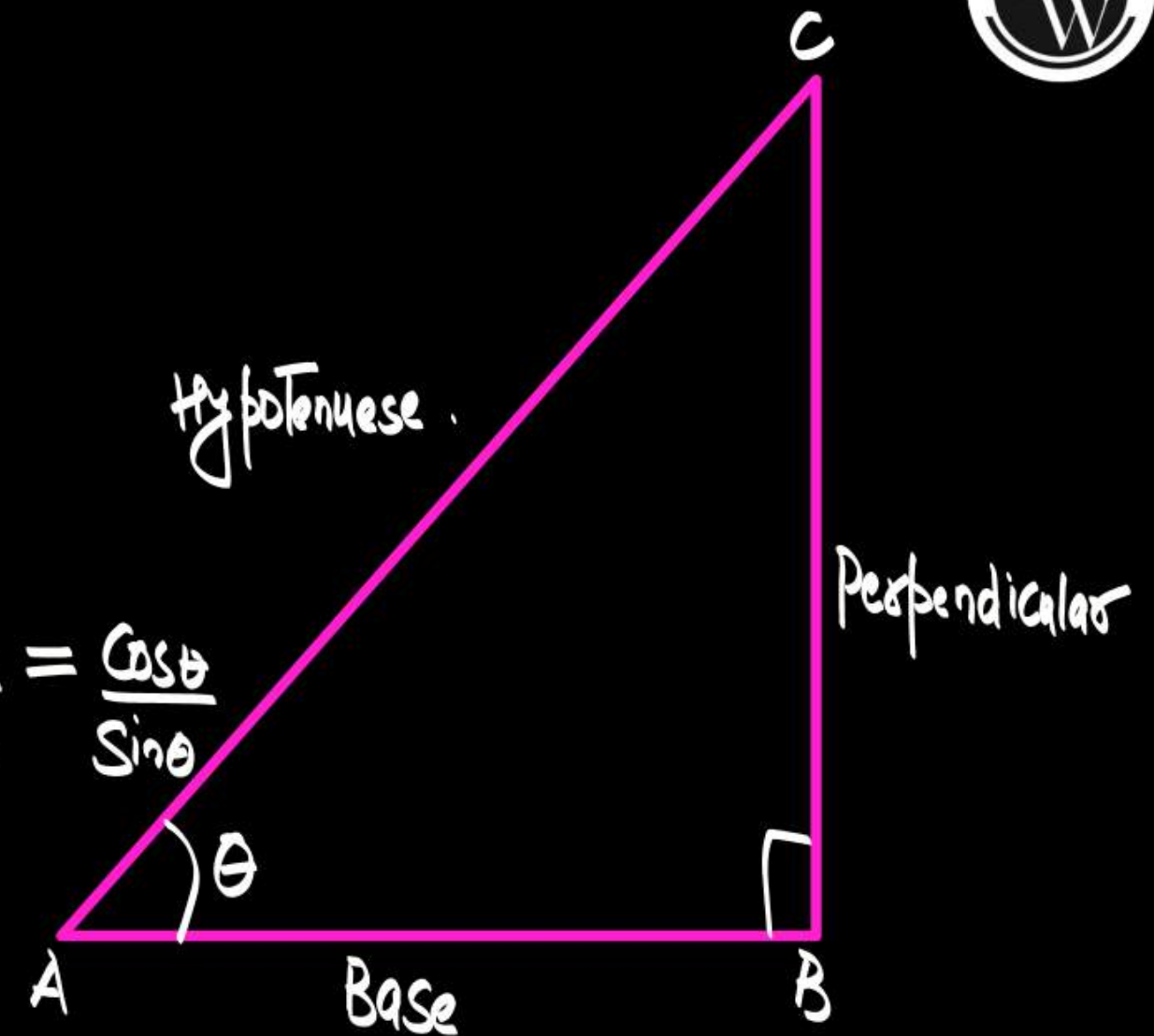
T-Ratios:

$$* \sin \theta = \frac{BC}{AC}, \cos \theta = \frac{AB}{AC}, \tan \theta = \frac{BC}{AB} = \frac{\sin \theta}{\cos \theta}$$

$$* \operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{AC}{BC}, \sec \theta = \frac{1}{\cos \theta} = \frac{AC}{AB}, \cot \theta = \frac{1}{\tan \theta} = \frac{AB}{BC} = \frac{\cos \theta}{\sin \theta}$$

$$* \sin^2 \theta + \cos^2 \theta = 1, 1 + \tan^2 \theta = \sec^2 \theta$$

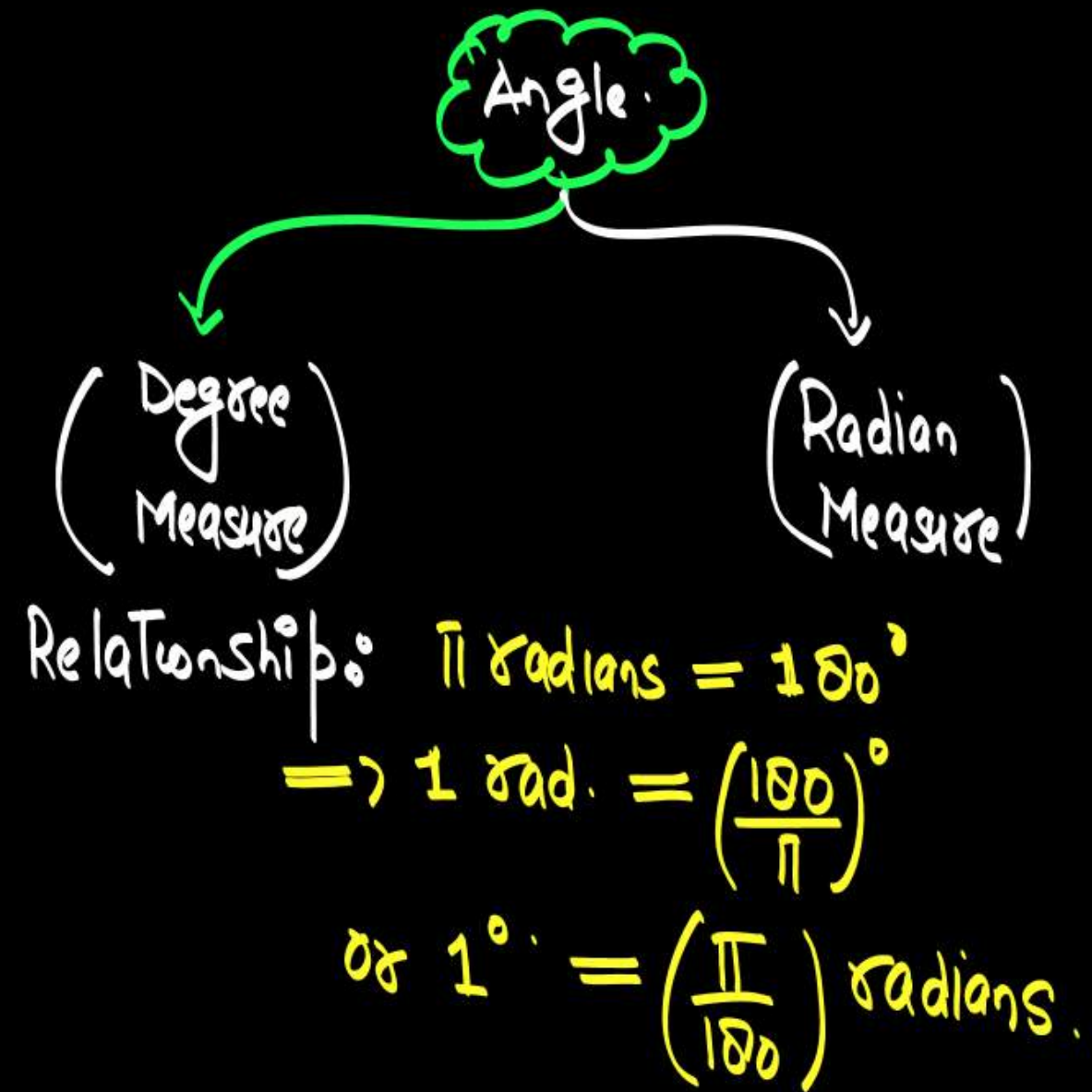
$$, 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta.$$



$$\frac{AB^2 + BC^2}{BC^2} = \frac{AC^2}{BC^2}$$

$$\cot^2 \theta + 1 = \operatorname{cosec}^2 \theta.$$

Angle	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	N.D.
$\operatorname{cosec} \theta$	N.D.	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	N.D.
$\cot \theta$	N.D.	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0



QUESTION



$$1 \text{ rad} = \left(\frac{180}{\pi}\right)^\circ$$

Find the degree measure of $-\frac{\pi}{3}$ and $+\frac{\pi}{9}$ respectively.

☒ A $20^\circ, -60^\circ$

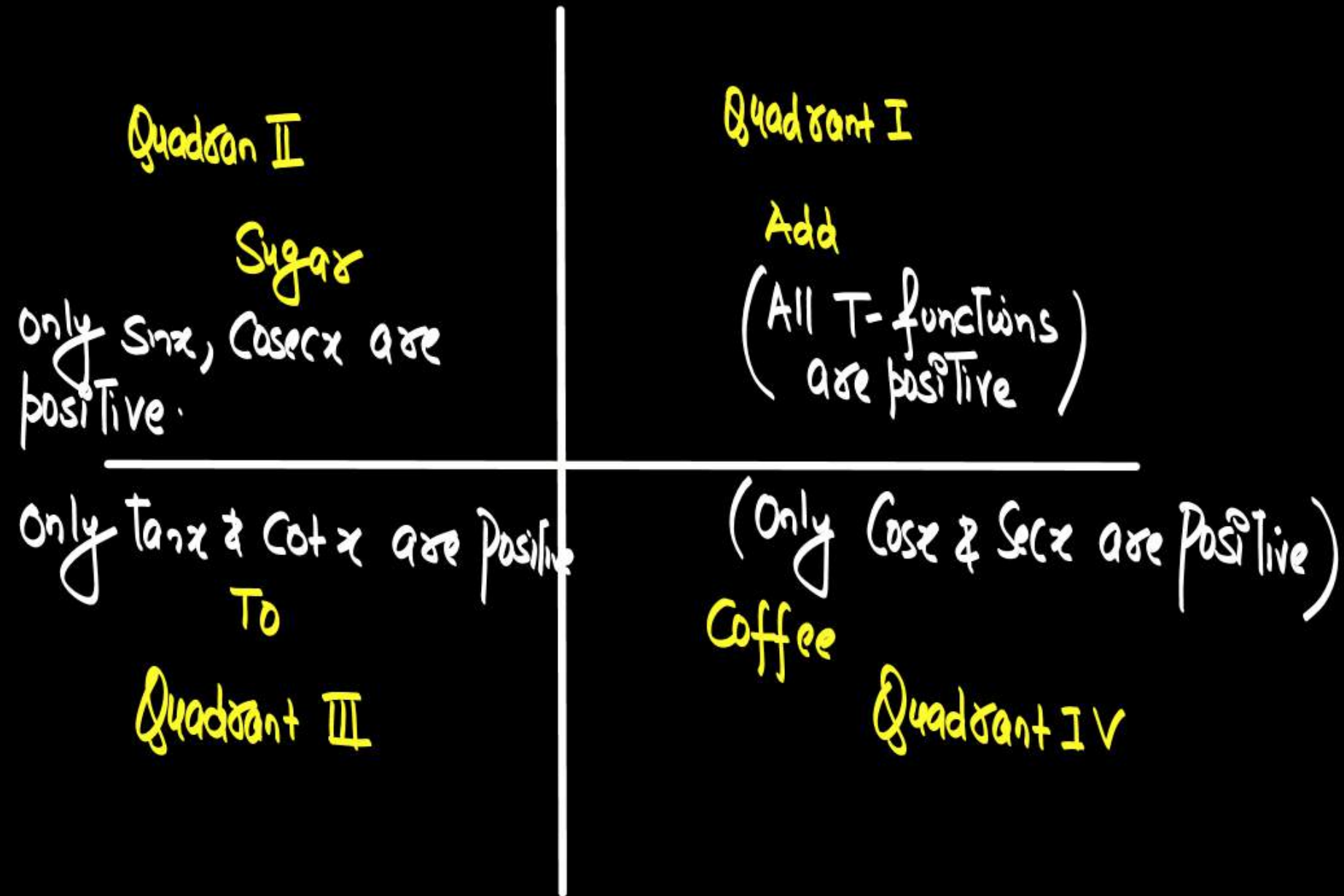
☐ B $30^\circ, -60^\circ$

☐ C $20^\circ, 40^\circ$

☐ D $40^\circ, 50^\circ$

$$-\frac{\pi}{3} = -\cancel{\frac{\pi}{3}} \times \frac{180}{\cancel{\pi}} = -60^\circ$$

$$\frac{\pi}{9} = \cancel{\frac{\pi}{9}} \times \frac{180}{\cancel{\pi}} = 20^\circ$$



QUESTION



If the arcs of the same lengths in two circles subtend angles 65° and 110° at the centre, find the ratio of their radii.

H/W

QUESTION



A wheel makes 360 revolutions in one minute. Through how many radians does it turn in one second?

hp/w

QUESTION

In a circle of diameter 40 cm, the length of a chord is 20 cm. Find the length of minor arc of the chord.

H/W

$$\theta = \frac{l}{r}$$



QUESTION



If in two circles, arcs of the same length subtend angles 60° and 75° at the centre, find the ratio of their radii.

H/w

QUESTION

The value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 179^\circ$ is

A 0 ✓

B 1 ✗

C -1 ✗

D $\frac{1}{\sqrt{2}}$ ✗

$$\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ \cos 91^\circ \dots \cos 179^\circ = 0$$

QUESTION

Find the values of other five trigonometric functions in Questions 1 to 5.

$\cos x = -\frac{1}{2}$ x lies in third quadrant.

(Sol) We know that $\sin^2 x + \cos^2 x = 1$.

$$\Rightarrow \sin^2 x + \left(-\frac{1}{2}\right)^2 = 1$$

$$\Rightarrow \sin^2 x + \frac{1}{4} = 1$$

$$\Rightarrow \sin^2 x = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\Rightarrow \sin x = \pm \frac{\sqrt{3}}{2}$$

Since x lies in the third quadrant,

$$\text{So } \sin x = -\frac{\sqrt{3}}{2}$$

Therefore the other T-functions are:

$$\sec x = -2, \csc x = -\frac{2}{\sqrt{3}}$$

$$\tan x = \frac{\left(-\frac{\sqrt{3}}{2}\right)}{\left(-\frac{1}{2}\right)} = \sqrt{3}, \cot x = \frac{1}{\sqrt{3}} \text{ Hence the result.}$$

QUESTION



Find the values of other five trigonometric functions in Questions 1 to 5.

$$\cot x = \frac{3}{4}, x \text{ lies in third quadrant.}$$

$$1 + \cot^2 x = \operatorname{cosec}^2 x$$

HP/W

Trigonometric Functions of Sum and Difference of Two Angles



We know that

$$\cos(x+y) = \cos x \cos y - \sin x \sin y \quad \text{--- (1)}$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y \quad \text{--- (2)}$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y \quad \text{--- (3)}$$

$$\sin(x-y) = \sin x \cos y - \cos x \sin y \quad \text{--- (4)}$$

Adding (1) & (2), we get;

$$\cos(x+y) + \cos(x-y) = \cancel{\cos x \cos y} - \cancel{\sin x \sin y} + \cancel{\cos x \cos y} + \cancel{\sin x \sin y}$$

$$\Rightarrow \boxed{\cos(x+y) + \cos(x-y) = 2 \cos x \cos y}$$

Subtracting (1) & (2), we get;

$$\cos(x+y) - \cos(x-y) = \cancel{\cos x \cos y} - \sin x \sin y - \cancel{\cos x \cos y} - \sin x \sin y$$

$$\Rightarrow \boxed{\cos(x+y) - \cos(x-y) = -2 \sin x \sin y}$$

Adding (3) & (4), we get;

$$\sin(x+y) + \sin(x-y) = \sin x \cos y + \cancel{\cos x \sin y} + \sin x \cos y - \cancel{\cos x \sin y}$$

$$\Rightarrow \boxed{\sin(x+y) + \sin(x-y) = 2 \sin x \cos y}$$

Subtracting (3) & (4), we get;

$$\sin(x+y) - \sin(x-y) = \cancel{\sin x \cos y} + \cos x \sin y - \cancel{\sin x \cos y} + \cos x \sin y$$

$$\Rightarrow \boxed{\sin(x+y) - \sin(x-y) = 2 \cos x \sin y}$$

* Put $x = \frac{\pi}{2}$ and $y = \theta$ in ① & ②, we get;

$$\cos\left(\frac{\pi}{2} + \theta\right) = \cos\frac{\pi}{2} \cos\theta - \sin\left(\frac{\pi}{2}\right) \sin\theta$$

$$\Rightarrow \boxed{\cos\left(\frac{\pi}{2} + \theta\right) = -\sin\theta}$$

Also $\cos\left(\frac{\pi}{2} - \theta\right) = \cos\left(\frac{\pi}{2}\right) \cos\theta + \sin\frac{\pi}{2} \sin\theta$

$$\Rightarrow \boxed{\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta}$$

$$* \sin\left(\frac{\pi}{2} + \theta\right) = \cos\theta, \sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$$

$$* \sin\left(\frac{3\pi}{2} + \theta\right) = -\cos\theta, \sin\left(\frac{3\pi}{2} - \theta\right) = -\cos\theta$$

$$* \sin(2\pi + \theta) = \sin\theta, \cos(2\pi + \theta) = \cos\theta$$

$$\tan(\pi + \theta) = \tan\theta$$

QUESTION



If none of the angles x , y and $(x + y)$ is an odd multiple of $\frac{\pi}{2}$, then

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

Proof: we know that

$$\tan(x + y) = \frac{\sin(x + y)}{\cos(x + y)} = \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y}$$

Dividing numerator and denominator by $\cos x \cos y$
we get;

$$\begin{aligned} \tan(x + y) &= \frac{\left(\frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y} \right)}{\left(\frac{\cos x \cos y - \sin x \sin y}{\cos x \cos y} \right)} \\ \Rightarrow \tan(x + y) &= \frac{\left(\frac{\cancel{\sin x \cos y} + \cancel{\cos x \sin y}}{\cancel{\cos x \cos y}} \right)}{\left(\frac{\cancel{\cos x \cos y} - \sin x \sin y}{\cancel{\cos x \cos y}} \right)} \end{aligned}$$

$$\tan(-\theta) = -\tan\theta$$

$$\Rightarrow \tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y} \quad \text{Hence proved.}$$

Remark: ① Here $\tan(x-y) = \tan[x+(-y)]$

$$\Rightarrow \tan(x-y) = \frac{\tan x + \tan(-y)}{1 - \tan x \tan(-y)}$$

$$\Rightarrow \tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\textcircled{2} \text{ Here } \tan 2x = \tan(x+x) = \frac{\tan x + \tan x}{1 - \tan x \tan x}$$

$$\Rightarrow \tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\textcircled{3} \text{ Here } \tan 3x = \tan(x+2x) = \frac{\tan x + \tan 2x}{1 - \tan x \tan 2x}$$

$$\Rightarrow \tan 3x = \frac{\tan x + \left(\frac{2 \tan x}{1 - \tan^2 x} \right)}{1 - \tan x \left(\frac{2 \tan x}{1 - \tan^2 x} \right)}$$

$$\Rightarrow \tan 3x = \frac{\tan x(1 - \tan^2 x) + 2\tan x}{\frac{1 - \tan^2 x - 2\tan^2 x}{1 - \tan^2 x}}$$

$$\Rightarrow \tan 3x = \frac{\tan x - \tan^3 x + 2\tan x}{1 - 3\tan^2 x}$$

$$\Rightarrow \tan 3x = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$$

Hence the result.

QUESTION



If none of the angles x , y and $(x + y)$ is a multiple of π , then

$$\cot(x + y) = \frac{\cot x \cot y - 1}{\cot y + \cot x} \quad \text{H/W}$$

QUESTION



(PYQJKB05E)

Prove that

$$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

Proof: We know that
 $\cos(x+y) = \cos x \cos y - \sin x \sin y$ — (1)
Replace y by x , we get;

$$\cos(x+x) = \cos x \cos x - \sin x \sin x$$

$$\Rightarrow \boxed{\cos 2x = \cos^2 x - \sin^2 x}$$

$$\begin{aligned} \Rightarrow \cos 2x &= \cos^2 x - (1 - \cos^2 x) \\ \Rightarrow \cos 2x &= \cos^2 x - 1 + \cos^2 x \\ \Rightarrow \boxed{\cos 2x = 2\cos^2 x - 1} \end{aligned}$$

$$\text{Also, } \cos 2x = \cos^2 x - \sin^2 x$$

$$\Rightarrow \cos 2x = 1 - \sin^2 x - \sin^2 x$$

$$\Rightarrow \boxed{\cos 2x = 1 - 2\sin^2 x}$$

Further $\cos 2x = \frac{\cos^2 x - \sin^2 x}{1}$
Dividing the numerator & denominator of the RHS by $\cos^2 x$, we get;

$$\Rightarrow \cos 2x = \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x}$$

$$\cos 2x = \frac{\left(\frac{\cos^2 x - \sin^2 x}{\cos^2 x} \right)}{\left(\frac{\cos^2 x + \sin^2 x}{\cos^2 x} \right)} = \frac{\left(\frac{\cancel{\cos^2 x} - \sin^2 x}{\cancel{\cos^2 x}} \right)}{\left(\frac{\cancel{\cos^2 x} + \sin^2 x}{\cancel{\cos^2 x}} \right)}$$

$$\Rightarrow \boxed{\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}}$$

QUESTION

Prove that

$$\sin 2x = 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x} \quad x \neq n\pi + \frac{\pi}{2} \text{ where } n \text{ is an integer}$$

Proof: We know that
 $\sin(x+y) = \sin x \cos y + \cos x \sin y$ (1)

Replace y by x in (1), we get:

$$\sin(x+x) = \sin x \cos x + \cos x \sin x$$

$$\Rightarrow \boxed{\sin 2x = 2 \sin x \cos x}$$

Here $\sin 2x = \frac{2 \sin x \cos x}{1} = \frac{2 \sin x \cos x}{\cos^2 x + \sin^2 x}$

Dividing the numerator & denominator of RHS by $\cos^2 x$, we get:

$$\sin 2x = \frac{\left(\frac{2 \cancel{\sin x} \cos x}{\cos^2 x} \right)}{\left(\frac{\cos^2 x + \sin^2 x}{\cos^2 x} \right)} = \frac{2 \tan x}{\frac{\cancel{\cos^2 x} + \sin^2 x}{\cancel{\cos^2 x}}} = \boxed{\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}}$$

Hence proved.

Prove the following: (PYQ J K BOSE)

(a) $\sin 3x = 3\sin x - 4\sin^3 x$

Proof: we know that

$$\sin(3x) = \sin(x+2x) = \sin x \cos 2x + \cos x \sin 2x$$

$$\Rightarrow \sin 3x = \sin x (1 - 2\sin^2 x) + \cos x (2\sin x \cos x)$$

$$\Rightarrow \sin 3x = \sin x - 2\sin^3 x + 2\sin x \cos^2 x$$

$$\Rightarrow \sin 3x = \sin x - 2\sin^3 x + 2\sin x (1 - \sin^2 x)$$

$$\Rightarrow \sin 3x = \sin x - 2\sin^3 x + 2\sin x - 2\sin^3 x$$

$$\Rightarrow \sin 3x = 3\sin x - 4\sin^3 x$$

Hence proved.

QUESTION

Prove the following:

(b) $\cos 3x = 4 \cos^3 x - 3 \cos x$

H/w

QUESTION



Prove the following:

(c) $\tan 3x = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$ if $3x \neq n\pi + \frac{\pi}{2}$ where $n \in \mathbb{Z}$

$$\tan 3x = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$$

Prove the following:

$$(i) \cos x + \cos y = 2 \cos \left(\frac{x+y}{2} \right) \cos \left(\frac{x-y}{2} \right)$$

$$(ii) \cos x - \cos y = -2 \sin \left(\frac{x+y}{2} \right) \sin \left(\frac{x-y}{2} \right)$$

$$(iii) \sin x + \sin y = 2 \sin \left(\frac{x+y}{2} \right) \cos \left(\frac{x-y}{2} \right)$$

$$(iv) \sin x - \sin y = 2 \cos \left(\frac{x+y}{2} \right) \sin \left(\frac{x-y}{2} \right)$$

QUESTION

If $\sin \theta + \cos \theta = 1$, then the value of $\sin 2\theta$ is equal to

- ☒ A 0
- ☐ B 1
- ☐ C -1
- ☐ D $\frac{1}{2}$

$$\begin{aligned} \text{(Sol)} \quad (\sin \theta + \cos \theta)^2 &= 1^2 \\ \Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta &= 1 \\ \Rightarrow 1 + \sin 2\theta &= 1 \\ \Rightarrow \sin 2\theta &= 0. \end{aligned}$$

QUESTION

The value of $\frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ} \sin$

$$\cos x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

A 2 ✗

B 1 ✗

C $\frac{\sqrt{3}}{2}$ ✓

D $\sqrt{3}$ ✗

$$\frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ} = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

QUESTION



$$\cos\left(\frac{\pi}{4}\right) = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Prove that

$$\cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) = \sqrt{2}\cos x$$

Proof: LHS = $\cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right)$

$$\Rightarrow \text{LHS} = \cos\left(\frac{\pi}{4}\right)\cos x - \cancel{\sin\left(\frac{\pi}{4}\right)\sin x} + \cos\left(\frac{\pi}{4}\right)\cos x + \cancel{\sin\left(\frac{\pi}{4}\right)\sin x}$$

$$\Rightarrow \text{LHS} = 2\cos\left(\frac{\pi}{4}\right)\cos x$$
$$\Rightarrow \text{LHS} = \frac{2}{\sqrt{2}}\cos x = \sqrt{2}\cos x$$
$$\Rightarrow \boxed{\text{LHS} = \sqrt{2}\cos x = \text{RHS}}$$

Hence proved.

QUESTION



Prove that *Imp.*

$$2\sin^2\frac{\pi}{6} + \underset{\text{H/W}}{\operatorname{cosec}^2\frac{7\pi}{6}}\cos^2\frac{\pi}{3} = \frac{3}{2}$$

QUESTION



Find the value of:

(i) $\sin 75^\circ$

$$\begin{aligned}\rightarrow \sin 75^\circ &= \sin(45^\circ + 30^\circ) \\ &= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}}\end{aligned}$$

QUESTION

Find the value of:
(ii) $\tan 15^\circ$

(Sol) we have $\tan 15^\circ = \tan(45^\circ - 30^\circ)$

$$\Rightarrow \tan 15^\circ = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} = \frac{\left(1 - \frac{1}{\sqrt{3}}\right)}{\left(1 + \frac{1}{\sqrt{3}}\right)}$$

$$\Rightarrow \tan 15^\circ = \frac{\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right)}{\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$\Rightarrow \tan 15^\circ = \frac{(\sqrt{3}-1)(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{(\sqrt{3}-1)^2}{\cancel{\sqrt{3}^2} - 1^2}$$

$$\Rightarrow \tan 15^\circ = \frac{\cancel{\sqrt{3}^2} + 1^2 - 2\sqrt{3}}{2} = \frac{4 - 2\sqrt{3}}{2}$$

$$\Rightarrow \tan 15^\circ = \frac{\cancel{2}(\cancel{2} - \sqrt{3})}{\cancel{2}}$$

$$\Rightarrow \tan 15^\circ = 2 - \sqrt{3}$$

Hence the result.

QUESTION



Prove that

$$\cos\left(\frac{\pi}{4} - x\right) \cos\left(\frac{\pi}{4} - y\right) - \sin\left(\frac{\pi}{4} - x\right) \sin\left(\frac{\pi}{4} - y\right) = \sin(x + y)$$

$$\cos\left[\left(\frac{\pi}{4} - x\right) + \left(\frac{\pi}{4} - y\right)\right] = \cos\left(\frac{\pi}{2} - (x + y)\right) = \sin(x + y)$$

QUESTION



Prove that

$$\text{LHS} = \cos\left(\frac{3\pi}{2} + x\right) \cos(2\pi + x) \left[\cot\left(\frac{3\pi}{2} - x\right) + \cancel{\cot} + \cot(2\pi + x) \right] = 1$$

Proof: $\text{LHS} = \sin x \cos x [\tan x + \cot x] = \sin x \cos x \left\{ \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \right\}$

$$\Rightarrow \text{LHS} = \cancel{\sin x \cos x} \left(\frac{\sin^2 x + \cos^2 x}{\cancel{\sin x \cos x}} \right)$$
$$\Rightarrow \text{LHS} = 1 = \text{RHS.}$$

QUESTION



Prove the following:

$$\sin(n+1)x \sin(n+2)x + \cos(n+1)x \cos(n+2)x = \cos x$$

H/W

QUESTION



Prove the following:

$$\cos\left(\frac{3\pi}{4} + x\right) - \sin\left(\frac{3\pi}{4} + x\right) = -\sqrt{2}\sin x$$

h/w

QUESTION



Prove the following:

$$\cot 4x(\sin 5x + \sin 3x) = \cot x (\sin 5x - \sin 3x)$$

Imp (HP/W).

QUESTION

Prove the following:

$$\frac{\sin x - \sin y}{\cos x + \cos y} = \tan \frac{x - y}{2}$$

h/w

QUESTION

Prove the following:

(PYQ JKBOSE)

$$\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$$

Proof: LHS = $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x}$

$$\Rightarrow \text{LHS} = \frac{(\cos 4x + \cos 2x) + \cos 3x}{(\sin 4x + \sin 2x) + \sin 3x}$$

$$* \sin x + \sin y = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$$

$$* \cos x + \cos y = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$$

$$\Rightarrow \text{LHS} = \frac{2 \cos\left(\frac{4x+2x}{2}\right) \cos\left(\frac{4x-2x}{2}\right) + \cos 3x}{2 \sin\left(\frac{4x+2x}{2}\right) \cos\left(\frac{4x-2x}{2}\right) + \sin 3x}$$

$$\Rightarrow \text{LHS} = \frac{2 \cos 3x \cos x + \cos 3x}{2 \sin 3x \cos x + \sin 3x} = \frac{\cos 3x (2 \cos x + 1)}{\sin 3x (2 \cos x + 1)}$$

$$\Rightarrow \boxed{\text{LHS} = \cot 3x = \text{RHS.}}$$

Hence proved.

QUESTION



(PYQ TK BOSE)

Prove the following:

$$\cot x \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x = 1$$

Proof: LHS = $\cot x \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x$

$$\Rightarrow \text{LHS} = \cot x \cot 2x - \cot 3x (\cot 2x + \cot x)$$

$$\Rightarrow \text{LHS} = \cot x \cot 2x - \cot(x+2x) (\cot 2x + \cot x)$$

$$\Rightarrow \text{LHS} = \cot x \cot 2x - \left(\frac{\cot x \cot 2x - 1}{\cot 2x + \cot x} \right) (\cot 2x + \cot x)$$

$$\Rightarrow \text{LHS} = \cancel{\cot x \cot 2x} - \cancel{\cot 2x \cot x} + 1$$

$$\Rightarrow \boxed{\text{LHS} = 1 = \text{RHS}}$$

Hence proved.

QUESTION



Prove that:

$$(\cos x + \cos y)^2 + (\sin x - \sin y)^2 = 4\cos^2 \frac{x+y}{2}$$

QUESTION



Prove that:

$$(\cos x - \cos y)^2 + (\sin x - \sin y)^2 = 4\sin^2 \frac{x-y}{2}$$

h/w

QUESTION

(PYQ-JKBOSE) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

Find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ in each of the following:

$\tan x = -\frac{4}{3}$, x in quadrant II

(Sol) We know that

$$\tan x = \frac{\tan\left[2\left(\frac{x}{2}\right)\right]}{1 - \tan^2\left(\frac{x}{2}\right)} = 2 \tan\left(\frac{x}{2}\right)$$

$$\Rightarrow -\frac{4}{3} = \frac{2t}{1-t^2}, \text{ where } t = \tan\left(\frac{x}{2}\right)$$

$$\Rightarrow -1(1-t^2) = 3t$$

$$\Rightarrow -2 + 2t^2 - 3t = 0$$

$$\Rightarrow 2t^2 - 3t - 2 = 0$$

$$\Rightarrow 2t^2 - 4t + t - 2 = 0$$

$$\Rightarrow 2t(t-2) + 1(t-2) = 0$$

$$\Rightarrow (2t+1)(t-2) = 0$$

$$\Rightarrow t = -\frac{1}{2} \text{ or } 2$$

$$\Rightarrow \tan\left(\frac{x}{2}\right) = -\frac{1}{2} \text{ or } 2$$

Since x lies in the 2nd Quadrant, so $\frac{x}{2}$ lies in the 1st Quadrant & thus $\tan\left(\frac{x}{2}\right) = 2$

$\therefore \sin\left(\frac{x}{2}\right) = \frac{2}{\sqrt{5}}$
 $\cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{5}}$

QUESTION



Find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ in each of the following:

$\cos x = -\frac{1}{3}$, x in quadrant II H/w

$$\cos x = \cos\left(2\left(\frac{x}{2}\right)\right) = 2\cos^2\left(\frac{x}{2}\right) - 1$$

QUESTION

Find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ in each of the following:

$\sin x = \frac{1}{4}$, x in quadrant II

H/W.

$$\sin x = \sin 2\left(\frac{x}{2}\right) = \frac{2 \tan\left(\frac{x}{2}\right)}{1 + \tan^2\left(\frac{x}{2}\right)}$$

Aksha De 

Sakib Ikhlag, Miao 

Mehak Jaan



Mehraj-4-Din Jan.

Thank you